

Rank Instability and Representation Repair for Automatic Modulation Classification under Structured Interference

Anonymous Author(s)

Abstract—Uniform AMC rankings can hide a more consequential failure mode: the ordering of compact spectral and high-capacity I/Q representations can change with interference conditions and fail to transfer across regimes. In source-disjoint simulations, the sealed family resolves an A5–A0 whole-configuration gain of 4.57 percentage points ([3.65, 5.49] percentage points), while the isolated teacher and route-count interventions remain unresolved. Marginally, the 39500-parameter A5 trails MCLDNN and IQFormer-inspired. Within the frozen campaign, however, a post-hoc SIR = −15 dB analysis reverses that ranking by 9.74 and 10.46 percentage points, respectively. A preregistered independently generated severe-held set then falsifies broad transport: pooled A5 gains are −3.12 percentage points against IQFormer-inspired and −12.45 percentage points against MCLDNN, while the frozen overlap–SIR envelope underperforms SIR-only or constant predictors. On the same held set, a prospectively designed received-I/Q sidecar raises macro-F1 by 4.37 percentage points ([2.96, 5.90] percentage points), with positive repair in all ten seeds and all eight SNR strata. It overtakes IQFormer-inspired but remains below MCLDNN. Probe, coverage, route-gating, and width-only controls further support a representation-information interpretation without proving it causally. The contribution is therefore a simulation-domain study of rank instability, prospective falsification, and representation repair, not a universally transportable operating boundary.

Index Terms—Automatic modulation classification, structured interference, rank instability, representation repair, domain shift, compact neural networks, interference overlap, vehicular Doppler.

Evidence status. The original sealed family resolves one positive whole-configuration contrast (A5 vs A0), while the isolated teacher and route-count contrasts and the clean-retention gate do not pass. Campaign-level severe-interference, envelope, Tier-2 repair, and capacity analyses are exploratory or post-hoc. Two later preregistered validations on an independently generated severe-held set respectively falsify broad transport and show strong received-I/Q repair; neither changes the sealed family.

I. INTRODUCTION

AUTOMATIC modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform’s modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. Structured interference and mobility can change which representation is preferred under a fixed compute budget. A marginal score then hides both the rank changes and the risk that a campaign-internal favorable region will fail under a new regime.

This paper asks whether compact spectral and high-capacity I/Q rankings remain stable across structured-interference conditions. It then tests whether an apparent campaign-internal boundary survives independent severe-domain validation. When that interpretation fails, we ask whether the degradation can be diagnosed and repaired at the representation level. We answer using a frozen source-disjoint simulation campaign, artifact-only analysis, and prospective interventions. The base model, VIMD-Net, maps a complex STFT to three allocation routes and uses a training-only simulation-component teacher. No sealed contrast confirms either element individually. The contribution is the rank-instability evidence, its prospective falsification, and the representation-repair chain.

The contributions are:

- a sealed comparison resolving an A5–A0 whole-configuration gain of 4.57 percentage points ([3.65, 5.49] percentage points), while the isolated teacher and route-count interventions remain unresolved;
- a complete SNR–SIR analysis showing that marginal model ordering is unstable inside the frozen campaign, including a post-hoc severe-condition reversal rather than a single global ranking;
- a preregistered robustness validation on 12,820 newly generated, source-disjoint severe-held examples that falsifies broad transport of both the campaign-internal corner and the frozen overlap–SIR envelope; and
- a prospective diagnosis-and-repair chain in which probe, exposure, route-gating, and width-only controls bound competing explanations, while a received-I/Q sidecar improves the independent severe-held set by 4.37 percentage points ([2.96, 5.90] percentage points).

The claim is deliberately bounded. Compact spectral and high-capacity I/Q representations exhibit condition-dependent rank instability inside the frozen campaign, but the apparent severe boundary does not transport uniformly. The strongest surviving intervention evidence is that restoring a small amount of received-I/Q information repairs substantial independent OOD degradation.

II. RELATED WORK AND CLAIM BOUNDARY

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated reparameterized causal convolutions, cross-scale kernel selection with IQ-imbalance and carrier-offset compensation, and

channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust, Cross-Domain, and Interference-Aware AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16], while overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation populate the 2024–2026 frontier [17]–[19]. Interference-tolerant recognition [16] and tensor-based malicious-interference recognition for MIMO [20] address structured interference directly. Our setting differs: we neither reconstruct sources nor seek universal interference robustness; we study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

C. Distinction: Rank Instability, Falsification, and Repair

Knowledge-guided and multimodal preprints continue these directions [21]–[24]. Our contribution is neither another average-dominance claim nor a universally valid selection surface. We study ranking instability, use a low-dimensional envelope as a diagnostic hypothesis, test its transfer under independent severe shift, and intervene on the exposed representation failure. This is a positioning claim, not an exhaustive systematic review.

III. SYSTEM MODEL AND VIMD CONFIGURATION

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa \mathcal{H}_u(u)[n] + w[n]), \quad (1)$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [25] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10 \log_{10}(E[|\mathcal{H}_s(s)|^2]/E[|\kappa \mathcal{H}_u(u)|^2])$ on the simulator's tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [26], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. The held-speed split probes deeper intra-window non-stationarity: $T_c \approx 0.423/f_D$ falls to about 0.31 ms at 250 km/h, below the 1024 μ s decision window.

Training pairs two independently impaired observations of the same immutable source; inference uses one view, so pairing is a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, swapping two admissible emitters can preserve the mixture while changing the requested label.

B. Three-Route Spectral Allocation

Let $Z = \text{STFT}(x) \in \mathbb{C}^{F \times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} [M_s, M_j, M_o] &= \text{softmax}([A_s, A_j, A_o]), \\ M_s + M_j + M_o &= 1. \end{aligned} \quad (2)$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ . The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \quad (3)$$

$$Z_j = (M_j + (1 - \lambda)M_o) \odot Z. \quad (4)$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range [0.05, 0.35]. It is added only to the modulation route; therefore the retained route weights remain nonnegative but sum to $1 + \rho$ rather than one, and the allocation is no longer an exact partition of the mixture power, i.e. $Z_m + Z_j \neq Z$ in general. This is the ratio-masking construction familiar from supervised source separation [27]–[30]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^* = [q_s - q_j]_+, \quad M_j^* = [q_j - q_s]_+, \quad (5)$$

$$M_o^* = q_u + 2 \min(q_s, q_j). \quad (6)$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

D. Training Objective

The total loss is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{mod}} + \alpha \mathcal{L}_{\text{teacher}} + \beta \mathcal{L}_{\text{jammer}} + \gamma \mathcal{L}_{\text{link}} \\ &\quad + \eta \mathcal{L}_{\text{contrast}} + \theta \mathcal{L}_{\text{orth}}, \end{aligned} \quad (7)$$

where $\mathcal{L}_{\text{mod}}$ is the modulation classification loss, $\mathcal{L}_{\text{teacher}}$ is a Jensen–Shannon [31] alignment between student and teacher

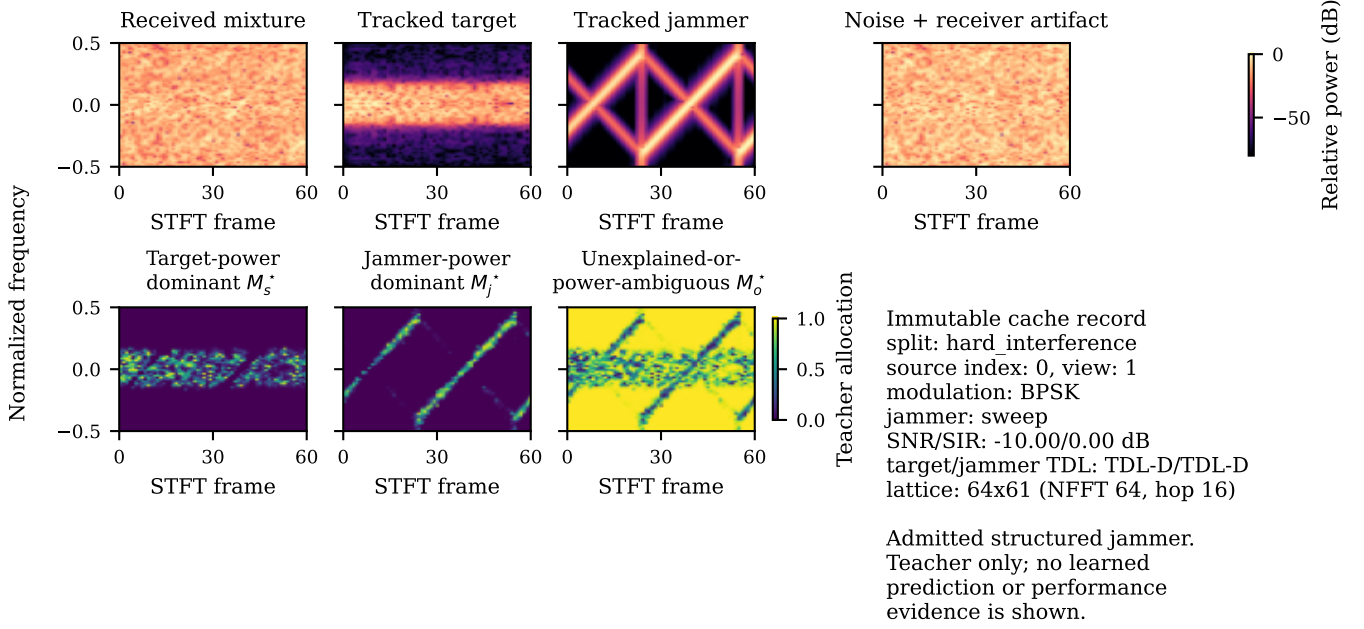


Fig. 1. A deterministic cache record and the fixed simulation-component teacher of Eqs. (5)–(6). Top row: received mixture, tracked target, tracked jammer, and noise-plus-artifact spectrograms; bottom row: the target-dominant, jammer-dominant, and uncertainty route masks. The teacher is a training-only construction from simulator-tracked components; it is unavailable at inference and is not performance evidence.

masks, $\mathcal{L}_{\text{jammer}}$ and $\mathcal{L}_{\text{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\text{contrast}}$ is a supervised-contrastive loss [32] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\text{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective, not a headline contribution.

E. Information Contract and Complexity

At inference every model receives only the mixture; simulator-tracked target, jammer, and SIR quantities are training-only or post-hoc audit variables. VIMD-Net uses 39500 parameters and 41.8 million reported MACs.

F. Implementation Details

Unless otherwise stated, all models share the same source-disjoint cache, data budget, optimization budget, stopping rule, and validation-only selection criterion; model-specific components follow each variant’s predefined configuration.

Input and front end. One example is a complex baseband window of 1024 samples ($1024 \mu\text{s}$ at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control. The spectral front end applies a two-sided STFT with $N_{\text{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospec-

tive capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4} , weight decay 10^{-2} , batch size 16, mixed precision, gradient-norm clipping at 5.0, and a shared 30-epoch budget. Auxiliary losses follow a fixed staged schedule, and the learning rate decays on a cosine to a floor of 0.05 times its initial value once the staged schedule is complete. The reported checkpoint is chosen solely by validation modulation cross-entropy under a common eligibility window and 8-epoch early-stopping rule. Optimizer settings, schedule, and selection rule are identical for every model; no comparator received an architecture-specific sweep, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 bundle but not isolated.

Seeds. The sealed campaign uses 10 algorithm seeds shared across all models, with seed-aligned initialization; matched-subset displays use the first five. Exact schedules, seed identifiers, teacher construction, and cache provenance are recorded in the reproduction package. Fig. 1 is a method illustration, not performance evidence.

IV. EVIDENCE PROTOCOL

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase,

and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans $\text{SIR} \in \{-15, -10, -5, 0\}$ dB, so $\text{SIR} = -15$ dB occurs only in the hard-interference split and $\text{SIR} = +5$ and $+10$ dB only in the in-distribution split. Each hard-interference SNR–SIR evaluation cell of Fig. 2 contains 131–181 source-disjoint test windows (mean 156; all ten classes, 5–26 per class), read from the frozen prediction cache. The envelope aggregates of Section V-C use a coarser partition of the same predictions (216–279 source-disjoint windows per held cell, mean 250). The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package but does not enter any sealed, envelope, or reported selector claim.

B. Comparator Fairness

Table I summarizes the comparators. All models are retrained under the same source-disjoint protocol; “IQFormer-inspired” and “MCLDNN” denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as an adaptation-oriented reference rather than a tuned upper bound; its 8.63 million parameters but only 155.3 million MACs reflect aggressive encoder downsampling and a large projection head, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator “IQFormer-insp.”; “A5” and “VIMD-Net” denote the same sealed configuration, and “CSSL” abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [33] we retrain under our protocol. Because this campaign retrains CSSL under the common supervised budget without executing the released self-supervised pretraining stage, it is reported as a supervised adaptation reference rather than a reproduction of the full released recipe.

As a public-benchmark comparator-fidelity check, the MCLDNN and IQFormer implementations were additionally evaluated on the public RML2016.10a benchmark [1] under the protocol of the official benchmark result table of the IQFormer release [34]: the 11-class set, the full -20 to 18 dB SNR grid, per-stratum 600/200/200 splits seeded at 233, no augmentation or input-length change. Over three algorithm seeds, all-SNR test accuracy is 58.55% for MCLDNN (seed range 58.1–59.4%) and 65.23% for IQFormer-inspired (64.8–65.8%; seed ranges, not confidence intervals); the reference-release values, averaged over the same twenty SNR bins,

TABLE I
COMPARATOR AND COMPLEXITY SUMMARY (HARD-INTERFERENCE SETTING).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

are 62.05% and 64.19%. IQFormer-inspired therefore closely matches the reference-release all-SNR accuracy (+1.04 pp), whereas MCLDNN remains moderately lower (−3.50 pp) while staying in the same broad all-SNR range. Per-SNR deviations of up to about 8 pp remain, so E1 is range-level comparator-implementation fidelity, not pointwise reproduction nor evidence for the simulator-domain envelope.

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A5 differs from A0 in allocation, teacher, auxiliary objectives, residual bypass, and capacity (about $4.7\times$ parameters), so A5–A0 is a whole-configuration effect.

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, overlap, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was preregistered separately and did not pass; clean behavior is therefore treated as a boundary to explain and repair, not as a preserved property of the sealed model. The later severe-held transfer and sidecar-repair validations were preregistered before inference on the independently generated held set. They remain post-campaign robustness evidence and do not alter the sealed family.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, an exploratory calibration rather than a sealed significance threshold. Because seeds are fixed blocks, the reported intervals are conditional on the realized seed set; a seed-randomization variant resampling the ten realized seeds with replacement keeps the A5–A0 contrast positive (95% percentile interval [4.02, 5.19] percentage

TABLE II
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES
ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE
POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	-0.76	1.08
Tri-route vs dual-route	0.44	-0.47	1.36

points) while the teacher-form and route-count contrasts span zero $([-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points). In the campaign’s severe row, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples. The SIR = -15 dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2.

V. RESULTS

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table II reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points, about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their simultaneous upper bounds are 1.08 and 1.36 percentage points, respectively. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three contrasts, the family gate did not pass. Because the A5–A0 interval is part of the same simultaneous family-wise band, its positive whole-configuration contrast remains a valid family-wise-controlled result even though the prespecified joint family gate does not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 4); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design’s power; we therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to any single component.

B. Campaign-Internal Rank Instability across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models. Figure 2 shows why no single campaign-level ranking is adequate: the sign changes with SNR and SIR, including a severe-row reversal. Section V-C summarizes this campaign structure without assuming that it transfers.

At SIR = -15 dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN $([7.52, 12.14])$ and 10.46 percentage points over IQFormer-inspired $([6.78, 14.29])$; these

two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks stabilize the reversal inside the campaign without converting the post-hoc analysis into confirmation. Section V-F prospectively tests a broad transport interpretation and rejects it on independently generated severe-held data.

C. Campaign-Diagnostic Overlap–SIR Envelope

The campaign-internal rank instability motivates a low-dimensional diagnostic surface. Rather than assuming a universal ranking, the surface summarizes how gain varies with target-support jammer overlap and SIR inside the frozen campaign. For reference b , we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b}o + \beta_{2,b} \text{SIR}, \quad (8)$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*. We avoid the more common “spectral occupancy” because o is not an occupied-bin or occupied-bandwidth fraction. The covariate is defined from the simulator-tracked target and jammer components rather than from the mixture. For each window we compute the STFT power of the tracked target and of the tracked jammer separately, order the time–frequency cells by descending target power, and take the target’s 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer’s total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t) \in \mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \quad (9)$$

Thus $o = 0.8$ means that 80% of the jammer’s energy lies within the target’s dominant support, not that the jammer fills 80% of the plane. The cell-level covariate is the mean of o over the windows of that cell.

Equation (9) is evaluated on a fixed analysis lattice, a Hann-windowed 128-point STFT with hop 32, deliberately independent of the 64-point VIMD front-end lattice of Section III-F: the front-end lattice is what the model sees, whereas the analysis lattice is a measurement instrument for the physical audit. The overlap is computed once per window when the cache is built, before any model is trained; it is stored in the sealed cache, checksummed in the cache manifest, and read unchanged by every model, split, and comparison in this paper. It is never an input feature of VIMD-Net or of any comparator, and it is not tuned against any model’s predictions.

The fit uses 32 aggregates, reconstructible from the protocol rather than asserted. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten classes. The `hard_interference` split contributes 4 SIR strata $(-15, -10, -5, 0 \text{ dB}) \times 4$ overlap quartiles = 16 cells, and `id_test` contributes 16 further cells over its five SIR strata; its lowest-overlap quartile is dominated by jammer-free windows ($o = 0$, SIR = 0 dB cache sentinel) and clears the window floor only at 0 dB, which fixes the per-stratum



Fig. 2. Campaign-internal post-hoc rank instability over all hard-interference SNR-SIR cells. Each entry is A5 minus the named I/Q reference in macro-F1 percentage points. Blue cells favor A5 and orange cells favor the reference; color and printed values encode the same sign. A star marks a positive unadjusted paired lower interval; Holm-surviving cells are reported in the text. The display does not imply transport outside the frozen campaign.

counts. The primary held set is the 80 finite-SIR interfered cells formed the same way from unseen jammer, unseen speed, held-out channel, and combined OOD (4×5 SIR strata $\times$ 4 quartiles = 80). Clean retention is deliberately outside this primary envelope because a jammer-free window has no physical finite SIR. Overlap and SIR are nearly orthogonal across the fitting cells (Pearson $r = -0.003$; -0.03 over all 113 cells), so strong linear collinearity between the two fitted covariates is not evident. We nevertheless interpret Eq. (8) as descriptive rather than causal; low pooled correlation bounds collinearity, not confounding.

Against IQFormer-inspired, the fitted overlap coefficient is 14.38 percentage points per unit overlap and the SIR coefficient is -1.036 percentage points per dB. Magnitude skill is scored against an explicitly defined trivial baseline: the constant predictor outputs the mean gain over the 32 fitting cells (-8.27 percentage points for IQFormer-inspired, -4.63 percentage points for MCLDNN) and is evaluated unchanged on the 80 held-regime cells, with $R^2_{\text{skill}} = 1 - \text{MSE}_{\text{envelope}}/\text{MSE}_{\text{const}}$. On that basis the envelope attains RMSE 6.23 percentage points against 8.37 percentage points for the constant predictor, for $R^2_{\text{skill}} = 0.45$ (Fig. 3); the MCLDNN envelope gives 8.10 percentage points against 9.63 percentage points ($R^2_{\text{skill}} = 0.29$). Adding the cell-level mean SNR as a third covariate lowers the held-regime RMSE only modestly (to 6.15 and 8.05 percentage points), so the two-variable model remains primary. A covariate ladder on the same cells (full table in the Supporting Material) isolates each predictor: SIR-only and overlap-only envelopes attain held RMSE 6.84 and 7.46 percentage points against IQFormer-inspired (8.71 and 8.68 percentage points against MCLDNN), versus 6.23 (8.10) percentage points for the two-variable model. Adding overlap to SIR reduces held-regime RMSE by 0.61 percentage points against both references, with $P(\Delta\text{MSE} > 0) = 0.94$ and 0.85 under a block-paired bootstrap, but both paired intervals include zero, so overlap remains an exploratory refinement beyond SIR. As a diagnostic rule, the envelope localizes the few held cells where the compact model becomes competitive: the sign rule

selects A5 in 5 of the 80 held cells, identifying all 3 hindsight-optimal cells (recall 3/3, precision 3/5) at 0.36 percentage points oracle regret, trading 0.22 percentage points realized macro-F1 for a 5.5% expected-MAC reduction versus always using IQFormer-inspired. The 80 held cells span SIR from -10 to $+10$ dB; the -15 -dB stratum occurs only in the hard-interference fitting split, so the held-regime transport metrics do not independently validate the break-even thresholds reported at -15 dB, although three held cells at $\text{SIR} \geq -10$ dB still favor A5 in hindsight; a subsequent preregistered check on an independently generated severe held set moreover does not support the -15 -dB transport (Section V-F). Regime-wise results are heterogeneous (regime table in the Supporting Material): the envelope substantially outperforms the regime-specific constant on the unseen-speed and held-channel shifts, but not on the unseen-jammer and combined-OD shifts. A hindsight oracle constant equal to the held-domain mean attains 6.04 and 6.81 percentage points; because it uses the held-domain mean, this pooled gap does not indicate missing within-regime structure. Sign agreement is 0.975 (0.950 for MCLDNN); with 0.963 (0.900) of held-regime gains negative, it is a secondary diagnostic only.

Solving Eq. (8) for zero gain gives break-even overlaps of 0.11 (vs. IQFormer-inspired) and 0.06 (vs. MCLDNN) at $\text{SIR} = -15$ dB, rising to 0.47 and 0.29 at $\text{SIR} = -10$ dB. The required overlap rises as interference becomes less severe, and these values are simulator-domain reference thresholds conditional on an overlap estimate, not deployment-ready control thresholds.

The preregistered mechanism expectation was non-increasing gain with overlap; the sealed directional test rejected that sign. Exploratory quintiles instead show monotonically increasing A5-CSSL gain, from a negative low-overlap region to a strongly positive high-overlap region; we return to this reversal in Section VI-C. The positive overlap direction is not specific to this adaptation reference: the fitted overlap coefficients against IQFormer-inspired and MCLDNN are also positive, at $+14.38$ and $+19.16$ percentage points per unit overlap, respectively.

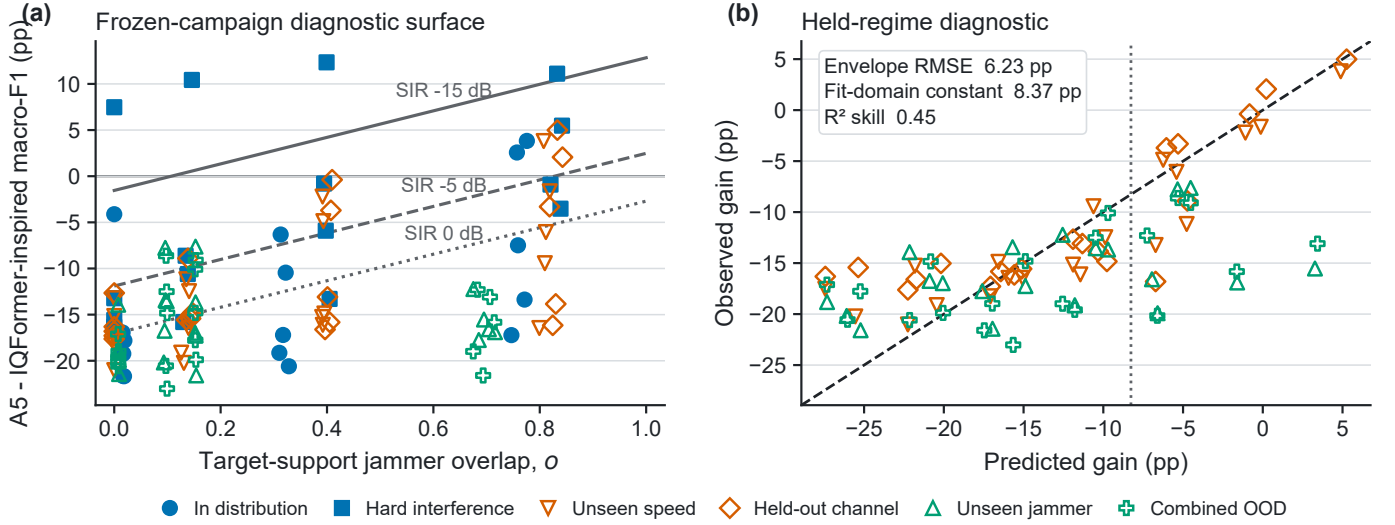


Fig. 3. Exploratory campaign-diagnostic overlap-SIR envelope against IQFormer-inspired. Panel (a) uses the target-support jammer overlap of Eq. (9), not occupied bandwidth. Filled blue markers are the 32 fit cells; open orange and green markers are the 80 held channel-side and interference-side cells. Panel (b) compares predicted and observed held-cell gain. RMSE is 6.23 percentage points versus 8.37 percentage points for the frozen fit-domain constant ($R^2_{\text{skill}} = 0.45$). The surface has limited held-regime skill inside the frozen campaign but is not assumed to transport to the independent severe-held regimes tested in Section V-F.

D. Complexity and Complementarity

A5 remains useful despite unstable ranking because its errors differ from those of the I/Q models. Late probability fusion improves over the stronger constituent by 4.71 percentage points with IQFormer-inspired and 5.70 percentage points with MCLDNN. The corresponding same-model seed-ensemble controls improve by only 1.49 and 1.69 percentage points, so the cross-model fusion gains exceed the ensemble controls by 3.22 and 4.01 percentage points. This is supporting evidence of complementary decisions, not marginal superiority.

Table I gives the corresponding parameter and MAC budgets; compactness is stated as a parameter and MACs property rather than an end-to-end inference-cost claim.

E. Clean-Boundary Diagnosis

The independent severe-held transfer failure makes representation robustness central to the interpretation. The earlier preregistered clean-boundary diagnosis independently tests whether the compact spectral representation discards information that becomes important under distribution shift. The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 5, where a star denotes a class whose jammer-free condition was not seen during training. Transfer failure occurs in 27/27 uncovered compact-spectral class-model cells (100%) versus 1/9 I/Q-reference cells (11%). The clean-retention gate did not pass. Figure 5 localizes the failure rather than hiding it in an aggregate.

We then followed a prospective decision chain. Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean

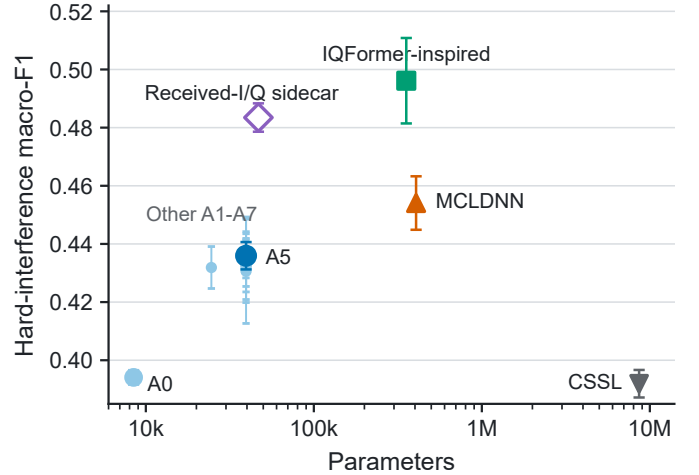


Fig. 4. Hard-interference macro-F1 versus parameter count on the matched five-seed subset. Error bars are ± 1 seed standard deviation. Blue marks denote the spectral family, green IQFormer-inspired, orange MCLDNN, gray CSSL, and purple the received-I/Q sidecar. Filled marks are sealed models; the open sidecar diamond is exploratory and outside the sealed family. The comparison is descriptive and regime-specific.

cross-validation; probe performance is the unweighted mean F1 over QPSK, 16QAM, and 64QAM (chance ≈ 0.1 on ten classes). The preregistered decision rule declares a representation *information-present, head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise, requiring *all* design-seed cells to exceed the threshold. For A5, five of the six design-seed cells (three algorithm seeds $\times$ the reconstructed jammer-free counterfactual and clean cross-validation probes) lie below the 0.25 threshold; the sole cell above it is borderline at 0.256, so we record the representation-limited side. The decision is unchanged at a threshold of 0.30 but flips at 0.20. The strong I/Q baselines exceed the threshold in

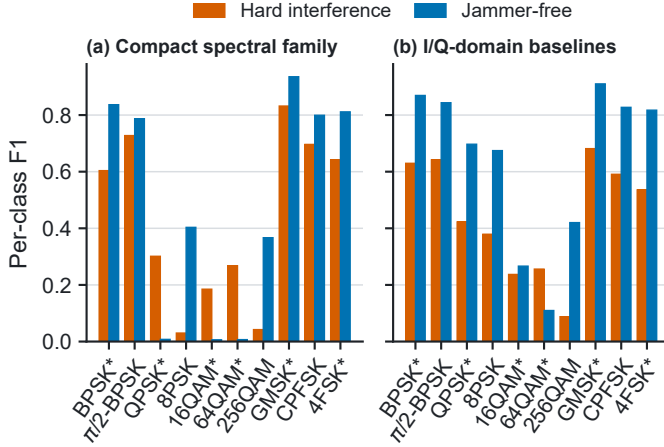


Fig. 5. Clean condition-transfer taxonomy. Bars show family-mean per-class F1 under hard interference (hatched orange) and jammer-free evaluation (blue) for the compact spectral family and I/Q-domain baselines. A star marks a class whose jammer-free condition was absent from training; it is not a significance marker. The same aggregation and scale are used in both panels.

18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, saturating near the modulated route; this arm is a negative control, not stand-alone evidence that a functional mask fallback was learned.

F. Representation Repair under Independent Severe Shift

Widening the spectral model without changing its representation domain first separates capacity from representation. On the matched five-seed subset, hard-interference macro-F1 rises from 0.436 at S to 0.455 at M and 0.468 at L. The widest tier still trails IQFormer-inspired on the marginal hard split by -2.84 percentage points ($[-4.30, -1.31]$) and on clean retention by 18.1 percentage points. Inside the frozen campaign it retains the SIR = -15 dB advantage, but width alone does not remove the representation boundary.

A preregistered independently generated severe-held set then tests broad transport. It contains fresh sources disjoint from the campaign pool, covers two severe regimes, and evaluates all frozen checkpoints unchanged. The campaign-internal corner does not transfer: pooled A5 gains are -3.12 percentage points against IQFormer-inspired and -12.45 percentage points against MCLDNN. Envelope sign accuracy falls to 0.578 and 0.406, while SIR-only or constant predictors beat the full surface. This experiment falsifies broad transport of both the corner and the -15 dB envelope.

The representation intervention adds a two-channel depthwise-separable received-I/Q branch and residually fuses a 24-dimensional embedding with the spectral path. It uses only the received mixture at inference; layer-level detail is in the Supporting Material. The total size rises from 39500 to 46794 parameters and to 43.1 million MACs, about 3% above A5. Inside the original campaign, it raises clean macro-F1 by 11.77 percentage points ($[11.06, 12.49]$) and hard macro-F1 by 4.37 percentage points ($[3.44, 5.31]$).

On the same independent severe-held set, the frozen sidecar raises pooled macro-F1 from 20.72% to 25.09%, a paired gain of 4.37 percentage points ($[2.96, 5.90]$). Repair is positive in all 10 seeds and all 8 SNR strata. The sidecar overtakes IQFormer-inspired (23.29%) but remains below MCLDNN (33.75%). This is strong representation repair, not uniform model superiority. Both severe-held validations remain exploratory post-campaign evidence outside the sealed family.

VI. DISCUSSION

A. What the Rank Instability Establishes

The stable conclusion is not that A5 owns a severe-interference region. Model ranking is condition- and regime-dependent. Marginal scores favor the strong I/Q references, the frozen campaign contains a severe reversal, and an independent regime reverses that result again. This pattern supports bounded model comparison, not a transportable deployment selector.

B. Why the Diagnostic Envelope Does Not Generalize Uniformly

Equation (8) has skill over 80 held cells inside the frozen campaign, but that skill is heterogeneous by regime. The incremental overlap effect beyond SIR is not separately resolved, and leave- -15 -dB-out behavior depends on the reference. The preregistered directional hypothesis also fails: gain increases rather than decreases with overlap inside the campaign. The surface is therefore a diagnostic hypothesis, not a causal equation or a universal switching rule. Its overlap variable additionally requires simulator-tracked target and jammer components and is not directly observable at an uncooperative receiver.

C. Prospective Falsification as Evidence

The independent severe-held experiment was frozen before inference and was allowed to reject the campaign interpretation. Its negative result is evidence about the boundary of the original finding, not a post-hoc exclusion. It also prevents the in-campaign RMSE and break-even references from being promoted to broad severe-domain transport claims.

D. Representation Repair

The intervention chain constrains, but does not prove, the explanation. Frozen probes point toward missing constellation information; coverage and route-gating controls reject simpler exposure and fallback explanations; the width ladder shows that spectral capacity alone is insufficient. The received-I/Q sidecar then repairs both the original clean boundary and

the independent severe degradation. Its remaining gap to MCLDNN limits the claim to repair, not uniform superiority or causal confirmation.

E. Vehicular Relevance and Deployment Boundary

At $f_c = 5.9$ GHz and the held-speed conditions, the preferred representation changes across mobility and interference regimes. The nominal coherence time at 180 and 250 km/h is about 42% and 30% of the 1024- μ s decision window, increasing intra-window nonstationarity. The independent failure cautions against fixing a switching rule from one simulator campaign. The lightweight sidecar is a more robust low-cost intervention than the frozen envelope is a deployment rule.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has eight explicit limits. First, all results are from source- disjoint TR 38.901 TDL-profile simulations. They are not measured over-the-air or complete MAC-/network-layer V2X evidence. Second, the original sealed family resolves only the A5–A0 whole-configuration contrast; the isolated teacher and route-count contrasts and the clean-retention gate do not pass. Third, the independent severe-held validation shows that the campaign-internal corner and the frozen envelope do not transport broadly. Fourth, the strong sidecar repair remains preregistered post-campaign robustness evidence outside the sealed family. Fifth, IQFormer-inspired and CSSL run to the shared 30-epoch limit in all ten seeds, so their comparisons retain a budget-sufficiency caveat; MCLDNN early-stops in every seed. Sixth, both strong I/Q comparators are unified-retraining references, and the public-benchmark check establishes range-level implementation fidelity rather than exact recipe reproduction. Seventh, the teacher and overlap covariate require simulator- tracked components. The overlap is not directly available at an uncooperative receiver, and its break-even values are descriptive references. Eighth, the probe, exposure, route-gating, width, and sidecar chain is consistent with a representation-information gap but does not establish a causal mechanism.

Reproducibility is enforced at the artifact boundary: each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256; every figure is rendered from a CSV with its own provenance manifest; and the resolved macros, figure-source CSVs, per-seed aggregates, configurations, and the full provenance manifest are released upon acceptance. The frozen run stores cache identity, checkpoints, source-aligned predictions, paired statistics, and execution records, preventing a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. CONCLUSION

Compact spectral and high-capacity I/Q AMC representations do not admit a single stable ranking under structured interference. The sealed campaign resolves a positive A5–A0 whole-configuration contrast but not the isolated teacher or route-count effects. Post-hoc analysis then reveals a severe-condition reversal inside the campaign. A preregistered independent validation falsifies broad transport of that corner:

A5 no longer leads the strong I/Q references, and the frozen overlap–SIR envelope does not transport its -15 dB advantage. The same held set provides a stronger representation-level result. The received-I/Q sidecar improves A5 by 4.37 percentage points ([2.96, 5.90] percentage points), with positive repair in all ten seeds and all eight SNR strata, while adding about 3% MACs. Together with the probe, exposure, route-gating, and width controls, the evidence is more consistent with a repairable representation-information deficit than with exposure or capacity alone. The central result is rank instability, prospective falsification, and representation repair inside a bounded simulation domain, not a universal operating boundary or uniform superiority.

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REFERENCES

- [1] T. J. O'Shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Engineering Applications of Neural Networks*, 2016, pp. 213–226.
- [2] F. Meng, P. Chen, L. Wu, and X. Wang, "Automatic modulation classification: A deep learning enabled approach," *IEEE Trans. Veh. Technol.*, vol. 67, no. 11, pp. 10 760–10 772, 2018.
- [3] Y. Tu, Y. Lin, C. Hou, and S. Mao, "Complex-valued networks for automatic modulation classification," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 10 085–10 089, 2020.
- [4] Z. Zhang, H. Luo, C. Wang, C. Gan, and Y. Xiang, "Automatic modulation classification using CNN-LSTM based dual-stream structure," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13 521–13 531, 2020.
- [5] J. Xu, C. Luo, G. Parr, and Y. Luo, "A spatiotemporal multi-channel learning framework for automatic modulation recognition," *IEEE Wireless Commun. Lett.*, vol. 9, no. 10, pp. 1629–1632, 2020.
- [6] J. Cai, F. Gan, X. Cao, and W. Liu, "Signal modulation classification based on the transformer network," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 3, pp. 1348–1357, 2022.
- [7] M. Shao, D. Li, S. Hong, J. Qi, and H. Sun, "IQFormer: A novel transformer-based model with multi-modality fusion for automatic modulation recognition," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 3, pp. 1623–1634, 2025.
- [8] Y. Wang, J. Yang, M. Liu, and G. Gui, "LightAMC: Lightweight automatic modulation classification via deep learning and compressive sensing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 3491–3495, 2020.
- [9] T. A. To, A. Argyriou, A. A. Puspitasari, S. L. Cotton, and B. M. Lee, "Efficient automatic modulation classification for next-generation wireless networks," *IEEE Trans. Green Commun. Netw.*, vol. 10, pp. 249–259, 2026.
- [10] N. Tang, X. Wang, F. Zhou, S. Tang, and Y. Lyu, "Reparameterization causal convolutional network for automatic modulation classification," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8576–8583, 2024.
- [11] B. Ren, K. C. Teh, H. An, and E. Gunawan, "OFDM modulation classification using Cross-SKNet with blind IQ imbalance and carrier frequency offset compensation," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8389–8403, 2024.
- [12] S. Huang, Y. Chen, J. He, S. Chang, and Z. Feng, "Channel-robust automatic modulation classification using companding spectral quotient cumulants," *IEEE Trans. Veh. Technol.*, vol. 73, no. 11, pp. 17 749–17 753, 2024.
- [13] S. Wang, H. Xing, C. Wang, H. Zhou, B. Hou, and L. Jiao, "SigDA: A superimposed domain adaptation framework for automatic modulation classification," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 13 159–13 172, 2024.

- [14] C. Xiao, S. Yang, Z. Feng, and L. Jiao, "MCLHN: Toward automatic modulation classification via masked contrastive learning with hard negatives," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 14 304–14 319, 2024.
- [15] M. Du, J. Pan, and D. Bi, "A contrastive learner for automatic modulation classification," *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 3575–3589, 2025.
- [16] X. Sun, C. Wu, Y. Li, Y. Li, T. Huang, J. Yu, and H. Zhang, "Toward interference-tolerant automatic modulation recognition via multi-stage feature extraction network," *IEEE Trans. Veh. Technol.*, vol. 74, no. 8, pp. 12 629–12 640, 2025.
- [17] G. Jajoo and P. Singh, "Modulation classification for overlapped signals using deep learning," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 3839–3851, 2024.
- [18] A. Faysal, M. Rostami, T. Boushine, R. G. Roshan, N. Muralidhar, Y.-D. Yao, and H. Wang, "DenoMAE2.0: Improving denoising masked autoencoders by classifying local patches for automatic modulation classification," *IEEE Trans. Commun.*, vol. 74, pp. 929–943, 2026.
- [19] W. Li, Z. Chen, T. Liu, H. Chen, and G. Xu, "Confidence-guided prototypical contrastive domain adaptation for cross-domain automatic modulation classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 6929–6941, 2026.
- [20] J. Zhang, X. Zhang, W. X. Zheng, Y. Chen, N. Zhao, J. Li, and M. Liu, "Modulation recognition via tensor analysis for MIMO systems with malicious interference," *IEEE Trans. Veh. Technol.*, vol. 75, no. 8, pp. 18 609–18 613, 2026.
- [21] S. Wang, C. Wang, H. Xing, H. Mo, L. Han, and L. Jiao, "DKDNet: Dual knowledge and data-driven network for cross-domain automatic modulation classification," arXiv preprint arXiv:2607.08031, 2026.
- [22] J. Wang, W. Xie, Y. Mu, X. Liu, Z. Zhao, and J. Zhang, "Mixture-of-experts transformer for automatic modulation recognition," arXiv preprint arXiv:2606.09085, 2026.
- [23] Z. Yang, Y. Luomei, Z. Liu, Z. Liu, and F. Xu, "ZoomSpec: A physics-guided coarse-to-fine framework for wideband spectrum sensing," arXiv preprint arXiv:2604.13568, 2026.
- [24] Y. Liu, M. Liu, W. Xie, X. Liu, W. Liu, Y. Sun, X. Qiu, C. Yuan, and J. Li, "RIS-MAE: A self-supervised modulation classification method based on raw IQ signals and masked autoencoder," arXiv preprint arXiv:2508.00274, 2025.
- [25] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project, Tech. Rep. TR 38.901, V18.0.0, 2024.
- [26] —, "Study on evaluation methodology of new vehicle-to-everything (V2X) use cases for LTE and NR," 3rd Generation Partnership Project, Tech. Rep. TR 37.885, V15.3.0, 2019.
- [27] A. Narayanan and D. Wang, "Ideal ratio mask estimation using deep neural networks for robust speech recognition," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2013, pp. 7092–7096.
- [28] Y. Wang, A. Narayanan, and D. Wang, "On training targets for supervised speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 22, no. 12, pp. 1849–1858, 2014.
- [29] H. Erdogan, J. R. Hershey, S. Watanabe, and J. Le Roux, "Phase-sensitive and recognition-boosted speech separation using deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2015, pp. 708–712.
- [30] D. S. Williamson, Y. Wang, and D. Wang, "Complex ratio masking for monaural speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24, no. 3, pp. 483–492, 2016.
- [31] J. Lin, "Divergence measures based on the Shannon entropy," *IEEE Trans. Inf. Theory*, vol. 37, no. 1, pp. 145–151, 1991.
- [32] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 18 661–18 673.
- [33] M. Du, J. Pan, and D. Bi, "CSSL-AMC-Pytorch," [Online]. Available: <https://github.com/duminyang20/CSSL-AMC-Pytorch>, 2025, commit 2fbc5b3, accessed Jul. 28, 2026.
- [34] M. Shao, D. Li, S. Hong, J. Qi, and H. Sun, "IQFormer," [Online]. Available: <https://github.com/WestdoorSad/IQFormer>, 2025, commit 7ee6ac9; accessed Jul. 28, 2026.